

BILKENT UNIVERSITY

PHYS 102 Lab Project Report

**Investigating the Strength of Electromagnets: Effects of
Windings and Current**



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PHYS 102 EXPERIMENT: Investigating the Strength of Electromagnets: Effects of Windings and Current

Objective:

This experiment aims to demonstrate the variables that affect the magnetic field. We will use a coil in our electromagnet experiment to observe how the magnetic field is impacted by current (I) and the number of windings (N). The magnetic field is directly proportional to the number of windings and current, as we will see in the experiment.

Introduction:

A magnetic field (B) is a vector field that represents the influence exerted by electric currents and magnetic materials. It is responsible for the force experienced by moving charges and magnetic dipoles. The strength of a magnetic field is measured in Tesla (T) in the SI system, where 1 Tesla is defined as 1 Weber per square meter (Wb/m^2). The magnetic field plays a crucial role in electromagnetism, governing the behavior of charged particles and forming the basis of technologies such as electromagnets, electric motors, and generators (Griffiths 196).

The permeability of free space (μ_0) or vacuum permeability is one of the basic physical constants of nature supposedly controlling the interaction between the magnetic field and the substance or vacuum itself (Jackson 180). It specifies how well a medium can support such an establishment of magnetic fields. In SI units, its value is given as:

$$\mu_0 = 4\pi \times 10^{-7} \text{N/A}^2$$

(Jackson 182)

This constant is the link between the magnetic field and the current that produces it and acts as a crucial parameter in the studies of electromagnetism. The permeability of various materials is different, leading to a different way of responding to magnetic fields, the ferromagnetic materials drastically increase the effect of the magnetic field compared to vacuum.

As seen in Figure 1, the coil is made up of a conductor twisted in a ring form. When current flows through a circular coil, it transforms into a dipole (South, North) and produces a magnetic field surrounding it, just like magnets do.

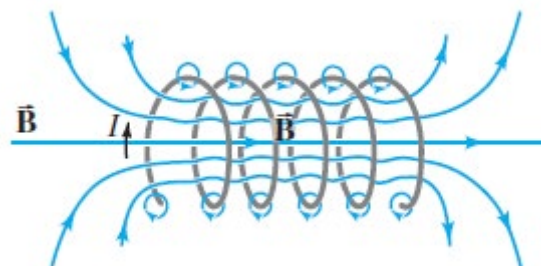


Figure1: Coil image displaying magnetic field lines

(Giancoli 819)

The magnetic field formula for a solenoid coil can be derived using Ampère's Law. In this formulation, B represents the magnetic field, l denotes the length of the wire, μ_0 is the permeability of free space, and I_{encl} corresponds to the enclosed current within the system (Giancoli 820). This relationship provides a theoretical foundation for examining the behavior of solenoid-generated fields in real-world applications by highlighting the magnetic field's reliance on basic electromagnetic principles.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$

(Giancoli 820)

After applying the necessary mathematical operations, the magnetic field of a solenoid coil can be expressed using the following formula:

$$B = \mu_0 I_{\text{encl}} \frac{N}{l}$$

where:

- B : Magnetic field
- N : Number of turns
- I : Current
- μ_0 : Permeability of free space constant
- l : Windings Length

We may compare the outcomes and assess the precision of the magnetic field generated by the coils using these calculations. Furthermore, we will test the strength of the electromagnets we have made using a Teslameter, sometimes referred to as a magnetometer or gauss meter. With the necessary measurements and the Teslameter, we will be able to assess the magnetic field and power generated by coils with varying numbers of windings and currents, while keeping the winding length constant.



Figure 2: Image of Teslameter

Equipments:

The materials to be used:

- Three solenoid coils with varying numbers of turns (20, 30, 40, and 50), all having the same winding length.
- Power supply
- Tesla meter (also referred to as a Gauss meter)

Procedure:

1. Coil Construction: To start off with, first, four coils will be constructed and will have 20, 30, 40, and 50 turns respectively in each of them. In each of the coils, they shall be wound to the same winding length, and the permeability of free space (μ_0) shall be considered constant too. With the electromagnets made from these coils, varying field strengths are expected from each of them depending on the number of turns.

2. Measuring the Magnetic Field of Each Coil (Stage 1): During this phase, four coils of 20, 30, 40 and 50 turns will be tested at constant current 4 amperes for actual measurement of the magnetic field from each coil with a Gauss meter.

- Step 1: Perform steps to connect each of the coils to the constant current of 4 amperes.
- Step 2: Gauss meter to measure the magnetic field each coil can generate.
- Step 3: Perform the measurements three times and calculate average values to increase accuracy.
- Step 4: Create records of measured values and compare them against theoretical predictions.

3. Investigating the Effect of Current on the Magnetic Field (Stage 2): At this stage, the effect of current on the magnetic field will be examined. In this respect, the 40-turn coil will be applied, testing at four different values of current (1 A, 3 A, 4 A, and 5 A).

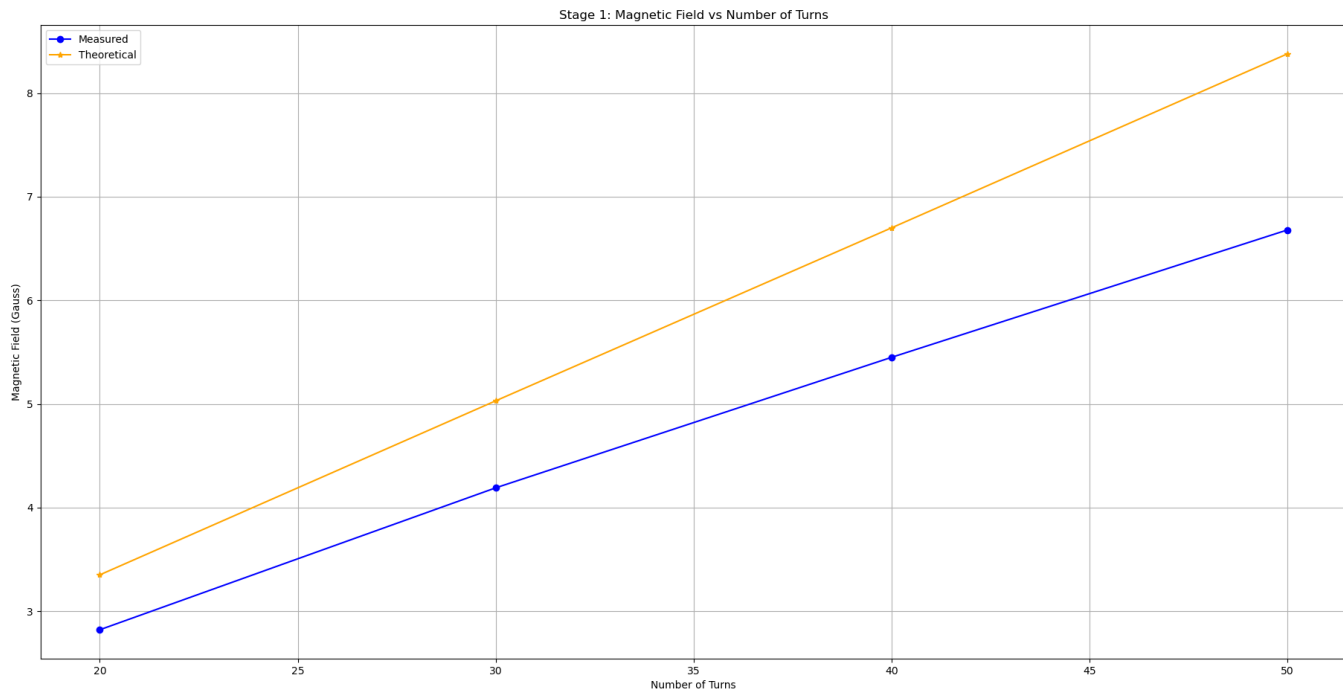
- Step 1: Select the coil of 40 turns.
- Step 2: Connect this coil to the four current values (1A, 3A, 4A, 5A) one by one.
- Step 3: Use the Gauss meter to measure the magnetic field generated by the coil for each of the set currents.
- Step 4: Repeat each measured average three times at each set current and record the findings in tables.
- Step 5: Compare the results with theoretical ones and derive error percentages.

4. Compare and Analyze Results: Data should be obtained at the end of both stages and compared, along with differences between theoretical values and measured values that would be analyzed using Ampère's law. Results will assist in understanding how both the number of turns and current affect the field strength.

Results:

	1. testing	2. testing	3. testing	Average Value	Theoretical Value	Error
20 Windings	2.85	2.82	2.80	2.82	3.35	15.8%
30 Windings	4.21	4.18	4.19	4.19	5.03	16.7%
40 Windings	5.47	5.46	5.43	5.45	6.70	18.7%
50 Windings	6.71	6.69	6.65	6.68	8.38	20.3%

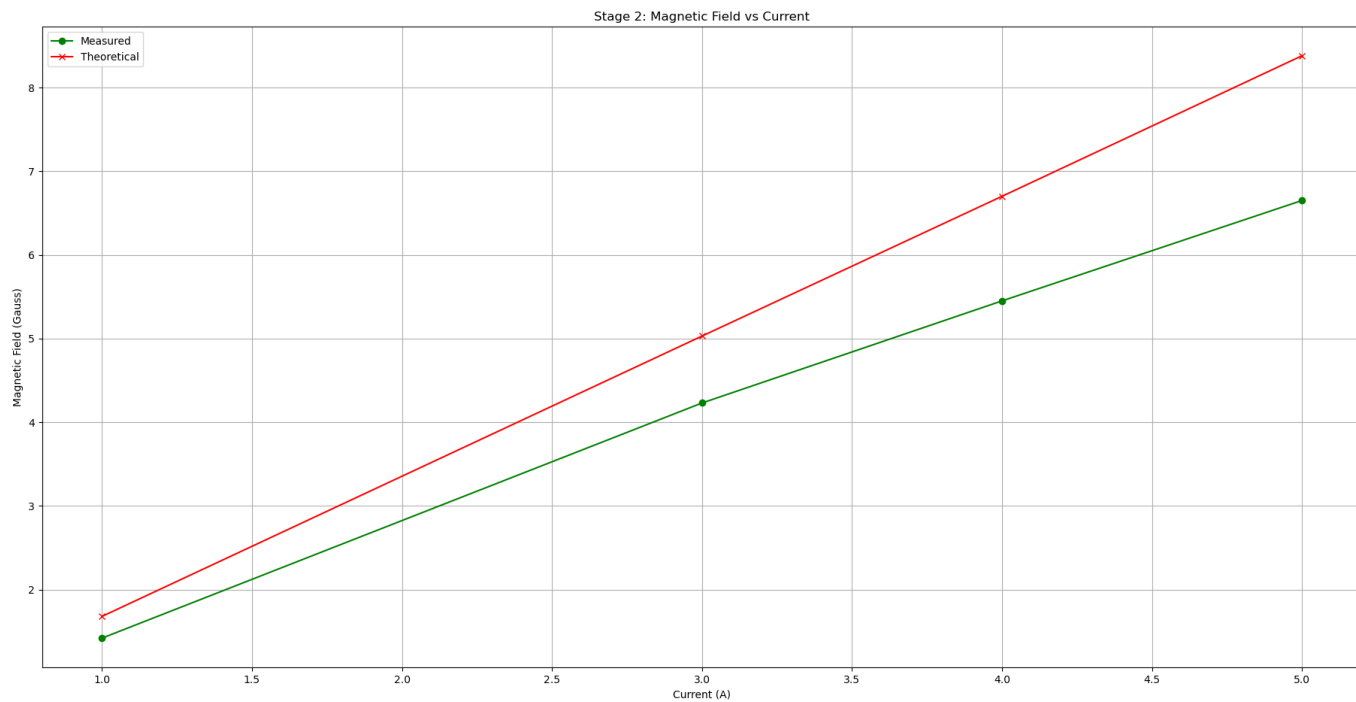
Table 1: Outcomes of 20, 30, 40, and 50 Windings



Plot of Table 1

	1. testing	2. testing	3. testing	Average Value	Theoretical Value	Error
1 Ampere	1.43	1.42	1.40	1.42	1.68	15.5%
3 Ampere	4.24	4.22	4.23	4.23	5.03	15.9%
4 Ampere	5.47	5.46	5.43	5.45	6.70	18.7%
5 Ampere	6.68	6.65	6.63	6.65	8.38	20.6%

Table2: Outcomes of 1,3,4 and 5 Ampere



Plot of Table 2

Conclusion:

In this experiment, we examined how the magnetic field strength of an electromagnet depends on two main variables: the number of turns on the coil and the amount of electric current flowing through it. According to Ampère's Law, the magnetic field inside a long solenoid is directly proportional to both the number of windings and the current. This relationship is expressed by the following formula:

$$B = \mu_0 \cdot \frac{N \cdot I}{l}$$

To test this formula, we constructed several coils by winding wires of different turn counts onto a 30-centimeter-long coil. Under different conditions, we measured the magnetic field each coil produced using a gaussmeter.

In the first stage of the experiment, the current was kept constant at 4 A, while the number of turns was gradually increased (20, 30, 40, and 50 turns). The measured average magnetic field values were recorded as 2.82 G, 4.19 G, 5.45 G, and 6.68 G, respectively. In the second stage, the number of turns was fixed at 40, and the current was varied (1 A, 3 A, 4 A, and 5 A). The corresponding measured average magnetic fields were 1.42 G, 4.23 G, 5.45 G, and 6.65 G. These results show that both parameters have a direct linear effect on the magnetic field strength.

When comparing the experimental data with the theoretical values, the percentage errors were found to range between 15.5% and 20.6%, which is acceptable under typical lab conditions. Interestingly, in test cases where the theoretical magnetic field should be the same (e.g., 30 turns – 4 A and 40 turns – 3 A), the measured value was lower in the high-current setup. This can be explained by the fact that higher current causes more heating in the wire, which increases its resistance and slightly reduces the actual current flowing through the coil. Therefore, even when the theoretical value is the same, setups with higher current often show greater experimental error.

The plots clearly reveal a linear relationship between magnetic field and both the number of turns and the current. The difference in slope between the theoretical and measured data lines visually highlights the systematic underperformance of the real system. These differences are mainly due to real-world factors such as imperfect coil winding, limitations in gaussmeter sensitivity, and Joule heating effects. Overall, the experiment strongly supported the theoretical model while also providing valuable insights into the physical limitations of electromagnets.

A youtube link to my experiment video:

<https://youtu.be/Nsslpiih3bw>

Sources:

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